CSCI 1300

Starting Computing

Hello!

My name is Amanda ☺

I am a current PhD student studying Computer Science and Cognitive Science.

Completed my undergraduate degree at Brown University in Computer Science and Cognitive Neuroscience

Research Interests:

- Human Computer Interaction, Human-AI Teaming, Multimodal Learning, Cognitive Neuroscience

Course Staff (1/2)

- Instructor: Amanda Hernandez
    - Office Hours:
        - Tuesdays 5-7PM MT, Zoom
        - Wednesdays 5-7PM MT, Zoom

- Graduate Teaching Assistant: Nolan Brady
    - PhD Student in Computer Science
    - Office Hours:
        - Tuesdays 12-2PM MT, Zoom
        - Thursdays 12-2PM MT, Zoom

- Course Assistant: Sashi Wolf
    - Undergraduate Student
    - Office Hours: TBA

Course Staff (2/2)

----- Also available for you!

- Instructor: Zachary Kaufman
  - Office Hours:
    - Mondays 9:30-11 am, ECCS 114A
    - Tuesdays 10:00-11 am, ECCS 114A
    - Fridays 9:30-11 am, ECCS 114A
- Graduate Teaching Assistant: Ishneet Chadha
  - Masters Student in Computer Science
  - Office Hours:
    - Mondays 11 am to 12 pm, Zoom
    - Tuesday 11:15 am to 12:15 pm, Zoom
    - Friday: 2:15 pm to 4:15 pm, Zoom

Computer Science

- CSCI1300 is a start to understanding computer science
  - for your own intellectual interest
  - for its enrichment of other fields
  - for its combination of scientific, engineering, art and design concepts and practices, and as a “mode of thought” – “computational thinking”
- IT, or information technology, including CS, is key to the “knowledge economy”
- Omnipresent in a breadth of various applications and fields

AI Dominates the News, Investments, …

A pattern for convocation speeches:

"First of all, AI. AI AI AI: AI, AI AI AI AI. When considering AI, it is important to remember AI, not to mention AI, AI, AI, and AI. AI AI AI AI."

"Moreover, in the coming year, AI. AI, and also AI, AI, and AI. AI AI AI; AI, AI AI AI AI AI. The outlook is AI, AI AI AI, AI. Our plans are AI, AI AI AI. Academic improvement through AI AI AI AI AI AI."

"In conclusion, AI AI AI AI AI AI AI AI AI AI AI AI AI AI."

Why Should *You* Study Computer Science?

- For fun and intellectual excitement
- Really exciting era is just beginning
  - CS still a young discipline, computers just starting to act intelligently
- Fundamental “mode of thought”
- Increasingly important component of all other fields
- Despite job market downturn, still are exciting and impactful jobs in established companies, start-ups, research labs, and academia – GAI is raising the bar on required skills
- While big tech hiring decreases, startups and adjacent fields are still developing, e.g., healthcare, pharma, fintech, e-commerce, climate tech

Welcome To CSCI 1300!

“You're going to love it here in Pelican Town.”

Why Stardew Valley?

- SDV was created by one person over multiple years, including programming, pixel art, music, and story when he was unable to get a CS job after his CS degree

- We hope that, similarly, your CS education inspires you to create something that you are passionate about

Who is CSCI1300 For?

- Students with varying levels of programming experience, including *NONE!*
    - however, CS1300 still requires a serious commitment
- Most students have little or no programming experience
- This is **not a weeding-out course**, but it is time-consuming
    - don’t worry!! Both TAs, CA, and Instructors are here to help you throughout the entire course!

CS1300 isn't About Getting the Correct "Answer"

- It's about the ***process***, not just the final product!
  - design
    - planning efficient, effective designs for program structure
    - investing upfront (e.g., reviewing materials) saves time in long run
  - implementation
    - coding incrementally
  - debugging
    - diagnosing and fixing bugs/errors in code effectively

Syllabus

- No “grading on a curve”: you do the work, you get the grade you deserve! Thus, A is by far the most common grade

- **Course components**
  - 4 weekly lectures
  - 1 weekly recitation (lab)
  - Weekly homework assignment (With a Part A and Part B)
  - 1 Midterm Exam (no final)
  - Final Project

- **Grade Scale**
  - 5% Lecture Attendance
  - 45% Homework (5pts for Part A, 10pts for Part B)
  - 50% Midterm Exam & Final Project

HW Specifications (1/2)

Part A

- Must be completed (and submitted) during lab time, otherwise it will be a 0
- If completed during lab time, it will be graded as normal out of 5 points by the autograder
- Will cover topics from Mon and Tues of each week
- If you are unable to attend recitation, let your TA know BEFORE recitation, and Part A will be due by end of day

Part B

- Due the Tuesday after lab at 11:59pm
- Can be worked on during lab after Part A is completed
- Graded by the autograder out of 10 points
- Will cover topics from all lectures of that week (and therefore you will not have been taught all topics necessary for Part B by lab time)

HW Specifications (2/2)

- Late assignments can be turned in up to 48 hours late for 20% off

- When you submit to Gradescope it will give you immediate feedback, which will include what output it actually got from your program, as well as what output it expected to get back

- Please plan to turn in a completed and correct assignment, however, if you did miss points in Gradescope you can resubmit for -1pt (for every re-submission)

- We intend for Part A to be completable during recitation without a rush, and Part B to be done as homework, although you’re welcome to work on it during recitation after Part A

Exam/Project

- Midterm exam will cover all topics up to that point, format TBD
- Planned to include extra credit questions

- Final project will be a multi-week project with a full outline and starter code
- Planned to include stretch goals for extra credit
- Will be SDV themed, although playing SDV is NOT required

Attendance

Required for both lecture and recitation

Lecture:

- Polls throughout the section
- HIT SUBMIT!!!!!!!!!
- Worth 5% of your final grade

Recitation:

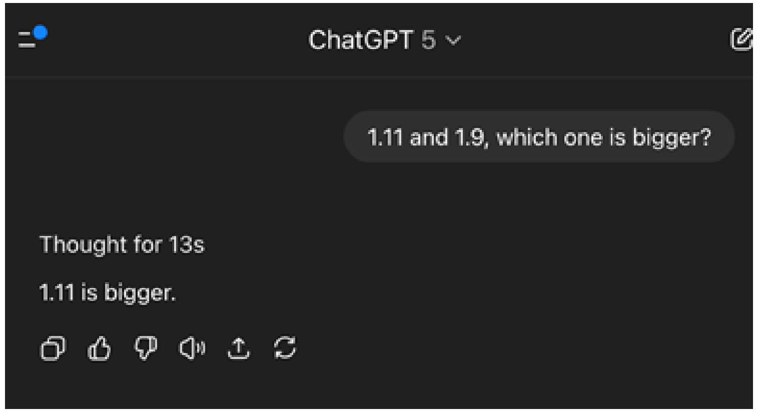
- Your TA will take attendance in whatever style they are most comfortable with
- Not graded directly, but Part A will get a 0 if you don't attend, even if you turn in Part A with a perfect score

Communications

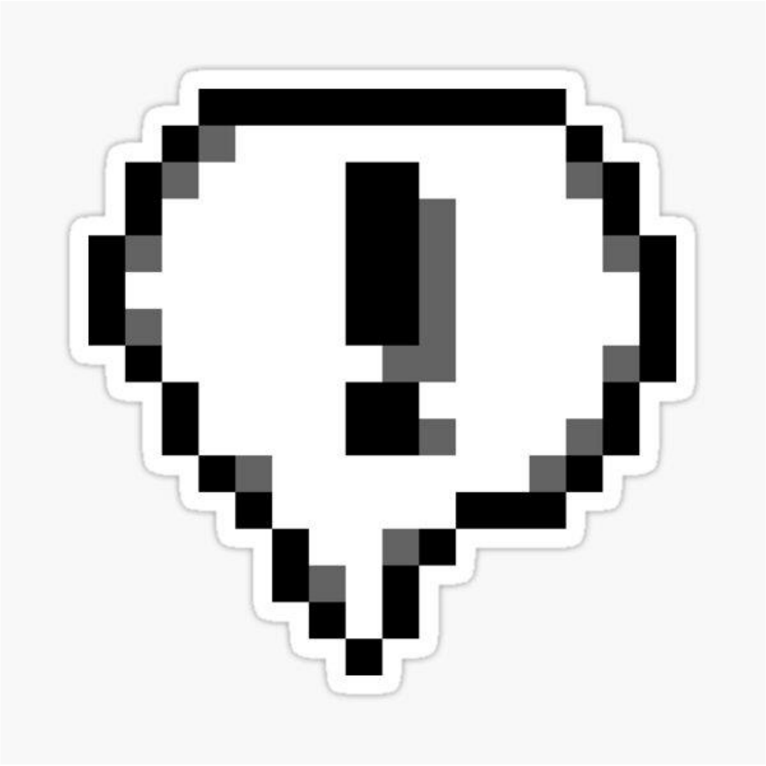
- Emails should be responded to within 24 hours (excluding weekends and holidays)
- If they aren’t responded to, ping again, don ’t feel bad!
- Discord messages should also be responded to within 24 hours
- Discord DMs should be for personal questions such as grade concerns, personal/medical problems, unable to attend class, etc.
- Discord messages within the server should be questions that other peers could benefit from getting answers for
- **Do not include code in server messages**

Generative AI (ChatGPT, Co-Pilot,…)

- Like any tech, can be used for good or bad, learn how to have a productive co-existence
- Unexpected by even the pros, incremental improvement this past year
- False confidence: GAI tends to present *everything* confidently as fact
    - GAI gurus are working on: guardrails, revealing sources, ingesting only curated training data, etc.
- MatLab, Wolfram Alpha can do the entire undergrad math curriculum, but…
    - learn the fundamentals so you have a sense of what it is doing, and what you can and can't trust
    - same way you can't rely on your peers in this course
- Your parents are paying you to learn, not just to get a good grade – need to master the fundamentals for future courses, jobs, etc.
    - if you cheat, you cheat yourself (and we will catch you!)



- Illegal collaboration is **not** worth the risk
    - start early and get help when you need it! Lots of resources available to help you succeed in this course



Syllabus or Logistics Questions?

Hope you’re excited for a great semester!

Announcements

- HW1 has been released tmrw
    - Part A is setting up your machine
        - Will be done during recitation tomorrow
    - Part B is due next Tuesday
- If applicable, submit your accommodation requests for the semester through [accommodate]
- If you haven’t, join the course Discord